

Scientists Borrow a Superpower from Wrinkly, Cancer-Proof Mole Rats--and Give It to Mice!

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Imagine an animal that's about the size of a mouse but can live longer than a pet cat up to 41 years! That's the **naked mole rat**, a wrinkly, pink rodent that scientists love to study. Why? Because these little guys almost never get cancer, and they stay healthy way into old age. Now, researchers have taken one of their secret weapons and given it to regular mice. The secret is a gooey molecule called **high molecular weight hyaluronic acid**, or HMW-HA (say "hi-a-lur-on-ic acid"). **Naked mole rats** have about ten times more of it than mice or humans. It's like a protective shield that fights **inflammation** (swelling that can damage your body) and stops **tumors** from growing. Scientists at the University of Rochester found the **gene** that makes this shield in mole rats. They carefully added that **gene** to mouse **DNA**. The result? The new mice had more HMW-HA, got fewer cancers, had less **inflammation** in their guts and other tissues, and lived about 4.4% longer than normal mice. That might not sound like a lot, but it's a huge deal. It proves that a longevity trick from one animal can work in another. Next, the team wants to see if this could help humans stay healthier as we age. "Our next goal is to transfer this benefit to humans," says lead researcher Vera Gorbunova. Who knew a wrinkly little rodent could hold clues to a longer, healthier life?

Word Treasure

naked mole rat -- A small, wrinkly, pink rodent that lives much longer than most similar animals and almost never gets cancer.

high molecular weight hyaluronic acid -- A thick, gooey molecule that forms a protective shield inside the body to fight swelling and stop tumor growth.

inflammation -- Swelling and redness inside the body that happens when you are hurt or sick, and can cause damage over time.

tumors -- Lumps or growths of abnormal cells that can sometimes turn into cancer.

gene -- A tiny piece of the DNA code that tells your body how to build something, like a protein.

DNA -- The chemical instruction booklet inside your cells that decides things like your eye color and how your body works.

Background you need

Naked mole rats are burrowing rodents from East Africa that live in large underground colonies, similar to ants or bees.

Hyaluronic acid is a natural substance in our skin, and many face creams include it to keep skin plump and hydrated.

Vera Gorbunova (a biologist at the University of Rochester, a university in the United States) has been studying why mole rats stay cancer-free for over 20 years.

Break it down

WHO: Naked mole rats, which have a special molecule called HMW-HA that protects them.

WHAT: Scientists transferred the gene for HMW-HA into mice, making them healthier and giving them longer lives.

WHERE: Research done at the University of Rochester in the USA.

WHY: To understand how animals resist aging and cancer, hoping to one day help humans stay healthier as they grow old.

Why it matters

Understanding how mole rats stay cancer-free could lead to medicines that keep you and your family healthier as you grow older.

Test what you remember

1. How long can a naked mole rat live?
 - (A) 5 years
 - (B) 15 years
 - (C) 41 years
 - (D) 100 years
2. What is HMW-HA?
 - (A) A type of food
 - (B) High molecular weight hyaluronic acid
 - (C) A rocket fuel
 - (D) A new video game
3. What does HMW-HA do in the body?
 - (A) Makes bones hard
 - (B) Fights inflammation and stops tumors
 - (C) Helps you run faster
 - (D) Makes you taller
4. What did scientists add to mouse DNA?
 - (A) A gene for making HMW-HA
 - (B) A gene for glowing in the dark
 - (C) A gene for bigger ears
 - (D) A gene for rainbow fur
5. How much longer did the modified mice live?
 - (A) 10 years
 - (B) 4.4% longer
 - (C) They lived forever
 - (D) 50% longer
6. What is the next goal according to Vera Gorbunova?
 - (A) To make mice live forever
 - (B) To transfer this benefit to humans
 - (C) To create a new species
 - (D) To win a Nobel Prize

Answer key (try the quiz first!): 1: C 2: B 3: B 4: A 5: B 6: B

Questions to think about

1. **Optimistic:** If this works in humans, it could lead to pills that slow aging and reduce cancer, giving people more healthy years.
2. **Cautious:** Genetically altering animals can have side effects; we must carefully test if it is safe for people.
3. **Scientific:** This is still early research in mice, and many anti-aging treatments that work in rodents fail later in human trials.

MY THOUGHTS (at least 20 words):